CS3203 SOFTWARE ENGINEERING PROJECT

Project Proposal

Privacy Preserving Collaborative Resource and Task Orchestrator for Research Environments

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PROJECT PROPOSAL

1. Title of the project : Privacy Preserving Collaborative Resource and Task Orchestrator for Research Environments

2. Overview of the project

We are building a Private SaaS (Single Tenant) platform deployed on isolated lab servers, ensuring better data sovereignty without third party leaks. Featuring a web dashboard, mobile app[2] and encrypted backend, the system uses AI to process research tasks, chats and equipment bookings into automated meeting summaries[3], conflict free schedules and real time notifications.

3. Objectives of the project

- Design and package a single tenant, containerized architecture that empowers academic labs to self host their own isolated, encrypted database, minimizing third party data vulnerability.
- Implement a responsive web and mobile friendly interface[2] for real time task tracking and end to end encrypted messaging between team members.
- Develop a resource locking mechanism preventing overlapping bookings for shared workstations, lab equipment, hardware and software assets with lab manager approval.
- Evaluate the system's impact on reducing admin time for researchers through automated summaries[3].

4. The Need for the project

Right now, it is really hard for academic research teams to manage their daily tasks while keeping their data private. Many academic teams store sensitive, pre publication findings on multitenant public cloud apps, creating severe third party security risks. We address this by offering a Private SaaS model: labs get the modern utility of a web platform, but run it entirely inside their own private infrastructure where external access is physically and logically blocked. When one academic research team uses simple spreadsheets to manage shared lab equipment, it usually leads to people double booking hardware and losing important notes[1]. We need a secure, centralized system that prioritizes this on premise privacy, so that researchers can protect their work and stop wasting time on manual admin tasks.

5. Scope of the project

Main User Roles:

Institute Admin – Manages teams, users, shared resources, and security settings.

Research Team Lead / PI – Manages projects, assigns members, and controls access.

Researcher / Team Member – Handles tasks, discussions, documents, notes, and bookings.

Resource Manager – Manages equipment, labs, rooms, and resource availability.

Main Functionalities:

- Task and Resource Orchestration: Using interactive Kanban boards for task management and an unified dashboard to prevent shared hardware booking conflicts.

- AI Context Engine: An independent backend service that processes our chat interactions, summarizes threads and extracts action items automatically[3].

6. Deliverables

- We are delivering our web platform as a pre configured Docker container, enabling research teams to instantly deploy a private, single tenant instance on local lab servers or private clouds. The interface features Kanban task boards, a resource booking dashboard and a collaborative rich text editor for documentation and meeting minutes. .
- The mobile app will provide push notifications and quick note capturing additional features.
- The backend will handle user authentication, project permissions, encrypted data storage, real time updates and automated context processing. The automated context engine will summarize discussions, extract action items and suggest task deadlines for users.

7. Overview of Existing Systems and Technology

Trello uses boards with cards, tracks tasks and gives updates away[4]. Slack is good for messaging, has channels and lets teams share files and work together[5]. Current systems mostly have websites with GUIs messaging tools, ways to track tasks and shared workspaces.
Unlike Slack and Trello, our system is privacy-first and self-hostable, combining task management with research resource scheduling and centralized documentation in one secure platform.

Our system merges these utilities using React.js for web dashboards and React Native for mobile push notifications[2]. The backend uses Spring Boot, PostgreSQL, and JWT for secure APIs and auth, combined with WebSockets and LangChain for real time updates and AI action item extraction[3]. To achieve our Single Tenant Private SaaS architecture, the entire full stack application and database are containerized via Docker for isolated deployment on private lab hardware.

8. References

[1] Reuben LIMS, "Why Spreadsheets Struggle as Laboratories Grow," reubendigital.co.uk. [Online]. Available: https://lims.reubendigital.co.uk/resources/blog/why-spreadsheets-struggle-as-laboratories-grow . [Accessed: July 3, 2026].

[2] React Native, "React Native: A framework for building native apps using React," reactnative.dev. [Online]. Available: https://reactnative.dev/ . [Accessed: July 3, 2026].

[3] LangChain AI, "LangChain Documentation," python.langchain.com. [Online]. Available: https://python.langchain.com/docs/get_started/introduction . [Accessed: July 3, 2026].

[4] Atlassian, “What is Trello?,” Trello. [Online]. Available: https://trello.com/tour. [Accessed: Jul. 3, 2026].

[5] Slack Technologies, “What is Slack?,” Slack. [Online]. Available: https://slack.com/intl/en-gb/help/articles/115004071768-What-is-Slack. [Accessed: Jul. 3, 2026].